

Task 9 – Descriptive Statistics Primer

Titanic Dataset | Veda Technologies – Data Analytics Internship

1. Project Overview

This project applies descriptive statistics to the Titanic dataset to summarize numerical variables and understand their central tendency, variability, percentiles, and distribution shape. The analysis is designed to be run in Python using Pandas, NumPy, and Matplotlib in Google Colab.

2. Objective

The objective is to calculate and interpret mean, median, mode, standard deviation, and percentiles for selected numeric columns, compare mean and median to identify possible skewness, and use visualizations to make the distribution easier to understand.

3. Tools and Technologies

Tool	Purpose
Python	Perform calculations and organize the analysis workflow
Pandas	Load, inspect, filter, and summarize tabular data
NumPy	Support numerical operations and conditional interpretation
Matplotlib	Create histograms and box plots
Google Colab	Run the notebook in a browser-based Python environment

4. Dataset and Selected Columns

The Titanic dataset contains passenger-level information. This notebook focuses on numeric fields that are commonly available in the dataset. Column names are checked before analysis so that the code uses only columns present in the uploaded file.

Column	Meaning
Age	Passenger age
Fare	Fare paid by the passenger
SibSp	Number of siblings or spouses aboard
Parch	Number of parents or children aboard
Pclass	Passenger ticket class

5. Statistical Measures

Mean: The arithmetic average of observed values.

Median: The middle value after sorting; it is generally less affected by extreme values than the mean.

Mode: The most frequently occurring value or values.

Standard deviation: Measures how spread out observations are around the mean.

Percentiles: Indicate the value below which a specified percentage of observations falls. The 25th and 75th percentiles define the middle 50%.

Interquartile range (IQR): The 75th percentile minus the 25th percentile.

Skewness: Describes asymmetry in a distribution. Positive values suggest a longer right tail; negative values suggest a longer left tail.

6. Analysis Workflow

- Load the Titanic CSV file into a Pandas DataFrame.
- Inspect shape, column names, data types, and missing values.
- Select available numeric columns for analysis.
- Calculate count, mean, median, mode, standard deviation, minimum, quartiles, and maximum.
- Calculate the interquartile range to describe the middle 50% of observations.
- Compare mean and median and calculate skewness.
- Create histograms and box plots to visualize distributions.
- Generate concise interpretations from the computed statistics.
- Export the summary table to CSV for reporting.

7. Visualizations

Histograms: Show the frequency distribution of selected numeric variables and help reveal concentration, spread, and skewness.

Box plots: Show the median, quartiles, overall spread, and potential outliers. Mean and median markers can be compared to help interpret the distribution.

8. Interpretation Guide

A noticeable difference between mean and median can suggest skewness, although it should be confirmed using the distribution plot and skewness value. A larger standard deviation indicates greater variability in the variable's original units. The 25th and 75th percentiles show the boundaries of the middle half of observations. Missing values can make the valid count lower than the total number of records.

9. Expected Observations to Verify

In many versions of the Titanic dataset, Age contains missing values and Fare may be positively skewed because a small number of passengers paid substantially higher fares. These are patterns to verify against the actual output, not assumed numerical findings. The final write-up should use the values and plots generated by the notebook.

10. Deliverables

- Jupyter Notebook with cell-by-cell analysis
- Summary statistics table for selected numeric columns
- Histograms and box plots
- Short interpretation of skewness and spread
- CSV export of the summary statistics

11. Key Learning

This task builds practical understanding of descriptive statistics and shows how central tendency, variability, percentiles, and skewness complement one another. Combining numerical summaries with visualizations helps explain what a dataset looks like instead of relying on a single statistic.

12. Conclusion

The project provides a repeatable workflow for summarizing numerical columns in the Titanic dataset. Using Pandas and NumPy for calculations and Matplotlib for visualization makes it possible to examine data distribution, variability,

missingness, and potential skewness. Final conclusions should be grounded in the notebook's computed results.

Organization: Veda Technologies

Program: Data Analytics Internship

Task: Task 9 – Descriptive Statistics Primer

Dataset: Titanic Dataset